MIL-L-10114A(CmlC)
27 June 1958
SUPERSEDING
MIL-L-10114(CmlC)
2 February 1950

MILITARY SPECIFICATION

LENS, CYLINCRICAL, LAMINATED, 3-PLY

(FOR PROTECTIVE MASKS)

1. SCOPE

1.1 This specification covers one type of protective mask, laminated glass lens.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawing, and publication of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

L-P-406 - Plastics, Organic: General Specifications, Test Methods
UU-P-313 - Paper, Lens Tissue
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Fiber.

MILITARY

MIL-B-117 - Bags, Interior, Packaging.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage.

FSC 4240

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DRAWINGS

CHEMICAL CORPS

C5-2-238 - Lens

PUBLICATIONS

CHEMICAL CORPS DIRECTIVE

No. 486 - Method for the Shadowgraph Inspection of Lens, Cylindrical, Laminated, 3-Ply.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

UNIFORM FREIGHT CLASSIFICATION

Ratings, Rules and Regulations

(Copies may be obtained from the Official Classification Committee, 1 Park Avenue at 33d Street, New York 16, New York.)

3. REQUIREMENTS

3.1 Material. - The lens shall be made from glass and transparent plastic binder.

3.2 Manufacture. - The lens shall be manufactured as specified in Drawing C5-2-238.

3.3 Transmission of light. - The lens shall transmit a minimum of 85 percent of the incident visible light when tested as specified in 4.3.1.

3.4 Prismatic effect. - The prismatic effect of the lens shall not exceed one-eighth diopter in the vertical meridian nor more than three-eights diopter in the horizontal meridan when tested as specified in 4.3.2.

3.5 Refractive power. - The refractive power of the lens shall not exceed plus or minus one-eighth diopter when tested as specified in 4.3.2.

3.6 Distortion. - The total distortion caused by irregularities in the lens shall not exceed three-eights diopter of lens power in any meridian when tested as specified in 4.3.3.

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3.7 Imperfections. - Imperfections in the lens shall not exceed the following limits when tested as specified in 4.3.4

3.7.1 Zone A (see fig. 1). - A maximum of two imperfections will be permitted provided the individual defective areas do not exceed that equivalent to a one-thirty-second-inch diameter circle and are separated by a minimum of one-fourth inch.

3.7.2 Zone B (see fig. 1). - A maximum of five imperfections will be permitted provided the individual defective areas do not exceed that equivalent to one-thirty-second-inch diameter circle and are separated by a minimum of one-fourth inch.

3.7.3 Zone C (see fig. 1). - Scratches in zone C which cannot be extended by manual stress will be permitted. Glass bubbles and shell-chipped or unbonded areas extending into zone C from the periphery not more than three-sixteenths of an inch, not more than one-half inch wide, and not deeper than 0.020 inch will be permitted, provided that they are separated by a minimum of 1 inch. Shell chips one-sixteenth inch or less in diameter will be permitted regardless of spacing. There shall be no vent chips, vent cracks, or sharp edges.

3.8 Separation (see 6.4).

3.8.1 Temperature change. - The lens plies shall not separate when tested as specified in 4.3.5.1.

3.8.2 Boiling. - Sections cut from the lens shall not show separation of the plies for more than one-fourth-inch from any edge, nor bubbles in the binder when tested as specified in 4.3.5.2.

3.8.3 Prolonged heating. - The lens plies shall show no separation when tested as specified in 4.3.5.3.

3.9 Impact. - The lens shall not fracture or crack when tested as specified in 4.3.6.1.

3.10 Spalling. - No fragment of glass exceeding 0.005 gram shall separate from the lens face opposite that receiving the impact when tested as specified in 4.3.6.2 (see 6.4).

3.11 Leakage. - The lens shall remain water-tight when tested as specified in 4.3.6.3.

3.12 Adhesion. - The adhesion of the glass layers to the binder shall be such that not more than 5 percent of the total binder shall be exposed when the lens is tested as specified in 4.3.7. (see 6.4).

4. QUALITY ASSURANCE PROVISIONS

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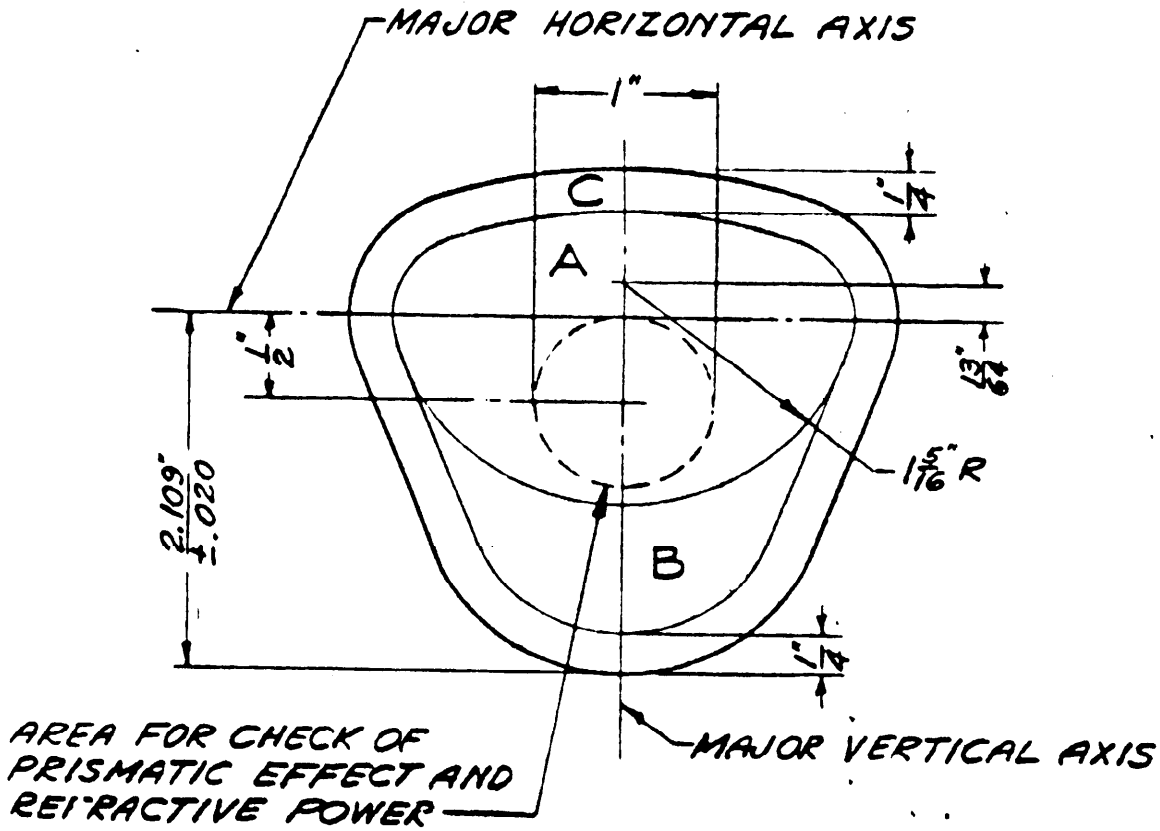


FIGURE 1

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4.1 Contractor's responsibility. - Unless othersise specified herein, the contractor is responsible for the examination and testing of the lenses prior to submission for acceptance.

4.2 Inspection provisions.

4.2.1 Materials. - The contractor shall present objective evidence as required by the Government representative that all materi als are in accordance with 3.1.

4.2.2 Lotting. - A lot shall consist of the lenses produced by one manufacturer under essentially the same manufacturing conditions from one lot of plastic. A sublot shall consist of that portion of the lot of lenses produced in twenty-four consecutive hours.

4.2.3 Sampling.

4.2.3.1 For nondestructive inspection. - Sampling shall be conducted in accordance with Standard MIL-STD-105.

4.2.3.2 For destructive testing.

4.2.3.2.1 First sublot. - From the first sublot of each lot, samples shall be selected at random for tests in accordance with table I:

Table I - Sampling for separation tests

Number of samples	Test	Retest	Test paragraph
20	10	10	4.3.5.1
20	10	10	4.3.5.2
20	10	10	4.3.5.3

4.2.3.2.2 All sublots. - From each sublot, including the first sublot, samples shall be selected at random for tests in accordance with table II:

Table II - Sampling for subsequent tests

Number of samples	Test	Retest	Test paragraph
30	15	15	4.3.6

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4.2.4 Examination.

4.2.4.1 Nondestructive. - The sample lenses from sublots or lots shall be inspected in accordance with 4.2.5 and Standard MIL-STD-105.

4.2.4.2 Destructive. - The sample lenses from sublots or lots shall be tested as specified in 4.3.

4.2.5 Classification of defects

4.2.5.1 Lens. - (Dwg. C5-2-238)

Categories and defects		Acceptance standards
Major A	AQL 1.0 percent defective	
1a.	Light transmission below minimum	4.3.1
2a.	Prismatic effect excessive	4.3.2
3a.	Refractive power out of limit	4.3.2
4a.	Distortion (zone A)	4.3.3
5a.	Imperfections (plastic defect-striation)zoneA)	4.3.4
6a.	Imperfections over limits (zone A)	4.3.4
Minor A	AQL 4.0 percent defective	
51a.	Distortion (zone B)	4.3.3
52a.	Imperfections (plastic defect-striation) (zone B);	4.3.4
53a.	Imperfections over limits (zone B)	4.3.4
Major	AQL 1.0 percent	
*1.	Cylindrical radius of curvature incorrect (zone C)	
2.	Profile incorrect	
3.	Etched line located incorrectly	
4.	Spotty discoloration (zone A)	
5.	Vent chips or vent cracks	
6.	Unbonded area (zone C)	
7.	Sharp edges or excessive chamfer	
8.	Outer ply overlapping in excess of 1/32 inch	
9.	Underply overlap	
10.	Scratches (zone C)	
11.	Damage	

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4.2.5.1 (continued)

Minor — AQL 4.0 percent defective

51. Glass bubbles (zone C)
52. Spotty discoloration (zone B)
53. Shell chip
54. Thickness incorrect

Incidental — AQL 10 percent defective

101. Contamination

* If the surface of the outer sheet deviates from the vertical axis, the lens shall not be classified as defective, providing the optical requirements are met.

4.2.5.2 Preparation for delivery - (section 5)

Categories and defects

Major — AQL 4.0 percent defective

1. Unit package quantity incorrect
2. Cushioning, when used, missing or incorrectly applied
3. Fiberboard tray, when used, provides insufficient clearance between lens
4. Unit package bag material incorrect
5. Unit package closure incorrect
6. Quantity per exterior container incorrect
7. Class and style of exterior container incorrect
8. Closure of exterior container incorrect
9. Strapping loose or missing
10. Loose package or pack
11. Unit packages or exterior container damaged
12. Marking incorrect, missing, or illegible

Minor — AQL 6.5 percent defective

51. Interior unit package carton incorrect
52. Strapping materials incorrect

4.3 Tests

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4.3.1 Transmission of light. - The transmission of light through the lens shall be determined in accordance with Method 3021 of Specification L-P-406.

4.3.2 Prismatic effect and refractive power. - The prismatic effect and refractive power of the lens shall be measured by means of a lensometer, vertometer, telescope, or equivalent. The measurements shall be taken over an area 1 inch in diameter, the center of which is on the vertical axis of the lens one-half inch below the intersection of the vertical and horizontal major axis (see fig. 1).

4.3.3 Distortion. - Distortion caused by irregularities in the lens shall be determined as specified in Directive No. 486 (see 6.5). In case of dispute the amount of distortion shall be determined by measuring the lens power of the disputed area by means of a lensometer or other suitable instrument, in which the negative power of the lens has been compensated by inserting in the light path an auxiliary lens having a plus one-fourth diopter cylindrical lens power. The refractive power of the compensated lens area shall be not greater than three-eighths diopter.

4.3.4 Imperfections. - Imperfections for which referee standards have been established shall be determined as specified in Directive No. 486 (see 6.5).

4.3.5 Separation. - Separation of the plies of the lens shall be determined as follows:

4.3.5.1 Temperature change. - Support the lens, by means of a wire screen, in a boiling water bath for 10 minutes and immediately thereafter place in a crushed ice bath containing enough water to cover the ice. Continually stir the bath for 10 minutes, then remove the lens and allow to stand for approximately 24 hours at room temperature. Examine the lens for separation of the plies.

4.3.5.2 Boiling. - Cut the lens in half along the vertical centerline and support by means of a wire screen in boiling water for 2 hours. Remove the lens from the bath and allow to stand at room temperature for approximately 24 hours. Examine the lens for separation of the plies.

4.3.5.3 Prolonged heating. - Place the lens in a closed container containing a sufficient amount of distilled water to insure saturated air for a minimum of 96 hours at 50°C. Heat the container and contents in a constant temperature oven at $50^{\circ} \pm 1^{\circ}C$ for 96 hours. Then remove the lens from the container and allow to stand 24 hours at room temperature. Examine the lens for separation of the plies.

4.3.6 Impact, spalling, and leakage. - The lens shall be subjected to the following tests, in order:

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4.3.6.1 Impact. - Mount the lens on a cylindrically formed, convex, wood surface, recessed under the lens, except for a supporting margin one-fourth-inch wide covered by a one-sixteenth-inch thick rubber gasket. Drop a hardened steel ball seven-eighths inch in diameter from a height of 15 inches, so that the impact on the lens will be applied perpendicularly to the tangent at a point on the vertical centerline one-half inch below the intersection of the vertical centerline with the horizontal major axis, on the convex surface of the lens. Examine the lens for fractures or cracks.

4.3.6.2 Spalling. - Repeat the test specified in 4.3.6.1 dropping the steel ball from a height of 30 inches. Individually weigh any fragments of glass that separate from the lower face of the lens.

4.3.6.3 Leakage. - Ascertain that the lens has been fractured. (In the event that any lens remains intact, return it to the apparatus and drop the ball from a sufficient height to fracture it.) Mount the lens in a suitable jig and subject it to a differential pressure of 27 inches of water for a minimum of one minute. Examine the lens for leakage of water.

4.3.7 Adhesion. - The lens shall be placed on a metal form having the same curvature as the lens and struck with a hammer until the glass is well broken on both faces of the lens, leaving the binder ply intact. After the loose particles of glass have been removed with a brush, the lens shall be examined for exposed binder.

4.4 Acceptance criteria.

4.4.1 Lot. - If two or three lenses fail when tested as specified in 4.3.5. the lot represented shall be retested as required using the lenses taken for this purpose. If more than three lenses fail in the test and retest combined, the lot shall be rejected.

4.4.2 Sublot. - If two or three lenses fail when tested as specified in 4.3.6 and 4.3.7, the sublot represented shall be retested using the lenses taken for this purpose. If more than three lenses fail in the tests and retests combined, the sublot shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A. - One hundred lenses shall be packaged on edge in a type I, class 1, style SFF, fiber box conforming to Specification PPP-B-636. The lenses shall be either individually interleaved with lens tissue paper conforming to type II of Specification UU-P-313, and overwrapped with same tissue material prior to

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placing in box; or the lens may be packaged by inserting each lens into a slot in a fiberboard tray in such a way that the lenses are not in contact with each other and not more than a one-eighth-inch width around the periphery of a lens contacts the tray before placing in box. Each box of 100 lenses shall be closed in accordance with the appendix to fiber box specification, and enclosed in a heat sealed bag conforming to type II, class e of Specification MIL-B-117. Staples shall not be used in box construction or box closure.

5.1.2 Level B.- Not applicable.

5.1.3 Level C.- Packaging shall be in accordance with the manufacture's commercial practice, when adequate to provide protection against damage to the items when packed as specified in 5.2.3.

5.2 Packing.

5.2.1 Level A.- Ten boxes of lenses, packaged as specified in 5.1.1, shall be packed in a class 2, style 2, 2-1/2, or 4 nailed wood box conforming to Specification PPP-B-621. The box shall be closed in accordance with Specification PPP-B-621 and strapped as specified in the apendix thereto, using galvanized strapping.

5.2.2 Level B.- Ten boxes of lenses, packaged as specified in 5.1.1, shall be packed in class 1, style 2, 2-1/2, or 4 nailed wood box conforming to Specification PPP-B-621. The box shall be closed in accordance with Specification PPP-B-621 and strapped as specified in the appendix thereto, using uncoated strapping.

5.2.3 Level C.- Lenses, packaged as specified in 5.1.3, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers used shall comply with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking. - In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with Standard MIL-STD-129. Packages and container shall be further marked to provide identification by sublot.

6. NOTES

6.1 Intended use.- This specification covers a lens intended to be used with M9 protective masks.

6.2 Ordering data.- Procurement documents should specify:

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a. Title, number, and date of this specification
b. Selection of applicable levels of packaging and packing.

6.3 Preliminary samples. - Compliance with the requirements for separation, adhesion, and spalling may be determined on preliminary samples, submitted at the time of initiating production, for approval of formulations for plastic layer and cement when used. The results of the tests will be considered informative only with respect to qualification of the lens formulation. In no case will such tests be considered as final acceptance tests.

6.4 Imperfections. - Details of imperfections with acceptable limits will be defined and controlled by Chemical Corps referee samples furnished as indicated in Directive No. 486.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Preparing activity:
Army - Chemical Corps

Other interest:
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